

Ryan Ting

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EDUCATION

Purdue University, West Lafayette, IN
Bachelor of Science in Industrial Engineering

May 2026

SKILLS

- Programming skills: Python, TensorFlow, OpenCV, Swift, CSS, C++, C, Arduino, MATLAB, JavaScript, Docker, GIT, Golang
- Software: Fusion360, Onshape, 3ds Max, Excel, PowerPoint, Word, SketchUp
- Engineering tools: 3D printer, Bandsaw, Drill Press, Milling Machine

RESEARCH

Purdue Mars Lab: *Undergraduate Researcher*

December 2021-May 2024

- Designed and prototyped a 5-DOF robotic gripper with rotating finger bases and multi-axis articulation for object manipulation.
- Developed CAD assemblies and mechanical architectures for a robotic arm integrating translational and rotational movement.
- Integrated motors and mechanical actuation systems into a functional prototype using custom 3D-printed components.
- Iterated through multiple redesign cycles to improve range of motion, manufacturability, structural reliability, and cost efficiency.
- Created detailed animations and mechanical demonstrations to validate kinematic behavior and assembly functionality.
- Designed and prototyped a multi-axis folding submarine fin mechanism with integrated rotational deployment and compact retraction capabilities for improved maneuverability and storage efficiency.

LEADERSHIP

Purdue ARC (Autonomous Robotics Club): Dog Copter *Project Manager*

Dec 2022 – May 2024

- Proposed and led development of a transforming drone-quadruped robotics platform integrating aerial and ground mobility systems.
- Recruited and coordinated a 20-member multidisciplinary engineering team spanning mechanical, electrical, and software.
- Established project architecture, timelines, and dependency tracking using Gantt charts and subsystem planning documentation.
- Developed MATLAB optimization tools evaluating 15+ design parameters for power-to-weight ratio and mechanical performance.
- Designed mechanical systems including folding propeller-leg attachment mechanisms and robotic structural assemblies.
- Built and tested miniature robotic dog prototypes utilizing servo motors and custom mechanical linkages.
- Organized subsystem leadership structure and technical review meetings to improve development coordination and integration.

PROJECTS/EXPERIENCE

Body-Controlled Search & Rescue Robot (Course Project)

April 2026 – May 2026

- Developed a real-time vision-based human-to-robot control system using gesture recognition.
- Built a face-based authorization system using OpenCV to enable secure access before allowing robot control.
- Developed a real-time hand tracking interface using MediaPipe, extracting hand landmarks to interpret gestures for directional control.
- Created a gesture-to-action mapping system that translates hand movements into servo steering and DC motor commands.
- Integrated vision-based control with Arduino hardware using Telemetry for real-time motor control in a live camera loop.

LSTM Stock Price Forecasting System: (Course Project)

April 2026 – April 2026

- Developed and trained stacked LSTM neural networks to forecast stock price trends from historical time-series data.
- Implemented data preprocessing, MinMax scaling, and train/test split evaluation on financial datasets.
- Built an iterative multi-step forecasting pipeline to generate future price predictions and evaluated performance using RMSE.
- Exported predictions to CSV and integrated outputs with an Arduino-based indicator system for decision logic.

Factory layout optimization algorithm

Sep 2025 – Sep 2025

- Developed a Python-based algorithm in IE 484 to optimize job shop floor layouts, including machine count and placement.
- Integrated techniques such as SPT for job shop, CRAFT for closeness rating and a custom clustering algorithm for plotting.
- Achieved the most efficient design in the class, requiring the fewest machines to meet production requirements.

Math mobile game

June 2025 – July 2025

- Developed a puzzle game inspired by the equation pyramid from The Devil's Plan (S2E11), featuring levels and custom game modes
- Built a robust algorithm to dynamically generate solvable number pyramids of configurable sizes.
- Used Kivy for the initial prototype, then rewrote the app in Swift due to iOS support issues.
- Deployed onto the App Store and achieved multiple downloads without marketing.

ETF Recommendation Bot

August 2024 – December 2024

- Developed a bot in Go to recommend ETFs, using an algorithm for finding similar ETFs based on customizable filters
- Self-taught Golang, gaining expertise in advanced topics like multi-level filter searching, sorting, and Excel file handling.
- Built comprehensive unit tests to validate functionality, ensuring 100% accuracy across all test cases.
- Designed an intuitive user interface for entering multiple ETF tickers or search parameters, with seamless export to Excel.

WORK EXPERIENCE

GAP Seasonal Sales associate

November 2024 – January 2025

- Assisted 50+ customers per shift while developing strong communication and problem-solving skills in a fast-paced environment.
- Supported inventory management operations including stocking, organization, and low-inventory monitoring.